Deepa Kale

806-778-3623 | dekale@ttu.edu | linkedin.com/in/deepa-kale | github.com/deepa-kale | Portfolio

Education

Texas Tech University, Lubbock, Texas May 2026
Bachelor of Science in Computer Science ***GPA:*** *3.5*

Academic Projects

Senior Capstone - TCP Traffic Anomaly Detection System | *Python, TensorFlow, Pandas, NumPy, Scikit-Learn* 2025
- Designed and implemented an end-to-end network traffic analysis pipeline that cleaned, normalized, and transformed raw TCP log data, extracted handshake features, and prepared structured datasets for anomaly detection
- Trained and optimized a CNN-based anomaly detection model over 100 epochs by tuning training hyperparameters and balancing normal and attack samples to reduce classification bias
- Achieved 94–96% training accuracy and 92–94% prediction accuracy on approximately 800,000 network traffic samples, evaluating performance using accuracy, precision, recall, and validation heatmaps
- Developed a modular command-line application that automated log ingestion, TCP handshake verification, CNN inference, and anomaly reporting, enabling users to analyze raw network logs within minutes and export anomaly summaries, anomaly distribution visualizations, and validation reports in CSV and image formats

Custom Apparel E-Commerce Platform | *React, JavaScript, Tailwind CSS, Node.js, Express.js, MongoDB* 2025
- Developed a full-stack e-commerce web application with product browsing, shopping cart functionality, inventory management, and backend API integration using React, Node.js, Express.js, and Tailwind CSS
- Designed RESTful API endpoints for product retrieval, cart operations, image handling, and frontend-backend communication using Express.js
- Structured backend logic using modular Express.js controllers to separate routing, business logic, and database interactions, improving maintainability and supporting future feature expansion
- Created MongoDB document schemas for products and transactions, designing flexible data models to support evolving product attributes while maintaining efficient querying and scalable data management

Hurricane Relief Resource Management Database (HEART) | *SQL, MySQL, Database Design* 2025
- Designed a relational database system to support emergency response operations by managing shelters, evacuees, volunteers, resources, and allocation records
- Created ER diagrams and normalized database schemas through Third Normal Form (3NF), defining primary keys, foreign keys, and constraints across entities to reduce redundancy and maintain data integrity
- Developed SQL queries using CTEs, JOINs, aggregations, filtering conditions, and subqueries to generate operational reports for shelter occupancy, volunteer assignments, resource distribution, and emergency response tracking

Leadership & Experience

Student Development and Engagement Lubbock, TX
Peer Mentor, Raider Ready Program & Tech Leadership Institute *January 2023 – June 2024*
- Mentored 40+ students through structured academic and career support, helping students navigate university resources and achieve personal goals
- Developed Excel-based tracking systems to monitor student attendance, mentoring progress, program participation, and SurveyMonkey student satisfaction data across Raider Ready and Tech Leadership Institute programs

Society of Women Engineers (SWE) Lubbock, TX
Outreach Committee Member *August 2023 – May 2026*
- Collaborated with engineering students to organize technical outreach events and increase participation in STEM initiatives

Technical Skills & Coursework

Coursework: Database Systems, Operating Systems, Applied Data Science, Data Structures & Algorithms, Design & Analysis of Algorithms, Object-Oriented Programming, Computer Organization & Assembly Language
Programming Languages: Python, Java, JavaScript, SQL, C/C++, HTML, CSS
Frameworks & Technologies: React, Node.js, Express.js, Tailwind CSS, REST APIs
Databases: MySQL, PostgreSQL, MongoDB, Relational Database Design, Query Optimization
Developer Tools: Git, GitHub, VS Code, Postman, JupyterLab, MATLAB, Google Cloud Platform